PROJECT INITIALIZATION AND PLANNING PHASE

Project Proposal (Proposed Solution) — CrimeVision

Date	1 November 2026
Team ID	[Enter Team ID]
Project Title	CrimeVision — Intelligent Multimodal Forensic Analysis Platform
Maximum Marks	5 Marks

Project Proposal (Proposed Solution) Report

CrimeVision is a prototype intelligent multimodal forensic analysis platform designed to help users organize digital evidence and perform an initial review of image, video, and audio files through a single application workflow. It combines evidence registration, local media-processing components, and saved analysis records to make findings easier to inspect and revisit. The system is intended to support investigators and analysts; its outputs require human verification and do not independently establish authenticity, guilt, or legal conclusions.

1. Project Overview

Objective. To develop a working prototype that accepts digital media as case evidence, runs supported analysis workflows, and presents the resulting information in an organized interface for review.

Scope. The current project includes a Flask-based web application, case and evidence management, file registration, image/object-detection workflow using OpenCV and Ultralytics/PyTorch, audio-transcription support through Whisper, saved analysis records, and a frontend for reviewing evidence and analysis results. Video can be registered with metadata and processed through supported video/frame workflows. Multimodal fusion and advanced forensic validation should be described as future work unless separately demonstrated in the running application.

2. Problem Statement

Description. Digital investigations may involve many images, videos, and audio files. Reviewing each file manually and switching between separate tools can be slow, repetitive, and difficult to organize. Findings may also be harder to compare or reopen when evidence and analysis records are kept in different places.

Impact. A fragmented review process can increase time spent locating relevant details, make case documentation less convenient, and raise the risk that useful observations are overlooked. Automated analysis can help prioritize material for review, but model errors and incomplete context mean that human examination remains necessary.

3. Proposed Solution

Approach. CrimeVision provides a unified workflow to register evidence against a case, process supported media using locally available models/libraries, and store analysis outputs so users can inspect the evidence and revisit saved results. The current prototype is configured for CPU-based PyTorch inference; it does not require a CUDA GPU for its present workflow.

Key Features

Feature	Description
Case and evidence management	Create and manage cases, register evidence items, and associate records with a case.
Multimedia evidence intake	Accept supported image, video, and audio uploads, subject to file-format and size validation.
Image/object detection	Use OpenCV and Ultralytics/PyTorch components for supported image analysis and object-detection workflows.
Audio transcription support	Use the installed Whisper workflow for speech transcription where supported and configured.
Saved analysis records	Store analysis outputs with case/evidence references so prior results can be retrieved.
Evidence review interface	Provide a web interface for browsing registered evidence and viewing saved analysis information.

Feature	Description
Local processing orientation	The current setup uses locally installed libraries/models for supported workflows; optional external services must be identified separately if used.

4. Expected Impact

CrimeVision is expected to make initial evidence review more organized, reduce repetitive tool switching, and make previously saved analysis easier to revisit. These are intended benefits of the prototype, not measured performance claims. Results must be interpreted in context and checked against original evidence.

5. Resource Requirements

Resource Type	Description	Current / Planned Allocation
Hardware	Development and test machine	Intel Core i5 12th Gen, 16 GB RAM, NVIDIA GTX 1650; current PyTorch environment is CPU-only.
Operating environment	Development platform	WSL 2 with Ubuntu; project environment managed through Python virtual environment.
Backend	Web application and API	Python, Flask, Flask-CORS.
Media / ML libraries	Image, video, object detection, audio	OpenCV, Ultralytics, PyTorch (CPU build), OpenAI Whisper, FFmpeg; availability depends on the installed environment.
Data storage	Case/evidence/analysis records	Project database and evidence files stored locally in the development environment.
Frontend	Browser interface	HTML/CSS/JavaScript; Three.js dependency may be used where configured.
External service	Optional language/model API	Groq SDK is installed and an API key may be configured; describe API-backed features only if actually invoked by the running application.

6. Limitations and Future Enhancements

Current limitations include CPU-only inference, dependence on model quality and supported file formats, and the need for human verification. A production forensic system would also require access controls, audit logging, secure evidence storage, documented chain-of-custody procedures, and formal validation. Potential future work includes stronger multimodal fusion, timeline generation, more detailed reporting, and improved forensic auditability. These should be presented as planned enhancements unless implemented and tested.

7. Conclusion

CrimeVision demonstrates a prototype workflow for organizing case evidence and using installed AI/media-processing components to assist initial review. Its value is in consolidating supported workflows and saved results in one interface, while keeping the investigator responsible for interpretation and verification.

Submission note: Replace [Enter Team ID] with the actual team ID. Confirm that every feature described as current is demonstrable in the submitted build.